

**TEZPUR UNIVERSITY**  
**Spring Semester End Term Examination, 2022**  
**CO 214: Computer Architecture and Organization**

Full Marks: 60

Time: 3 Hrs

*Figures in the right hand indicate marks for the individual question.*

**Section-I**  
**(40 Marks)**

1. Answer any two of the following:

- a. What is Isolated I/O? What are the additional requirements to support isolated I/O compared to memory mapped I/O? 2 x 4
- b. Explain the role of the Status Register in an I/O interface.
- c. Explain the 'Busy Wait' condition that occur in Program Controlled I/O and its effects.

2. a. What are the similarities and differences between an interrupt and a subroutine call? 4

Or

- b. Explain the mechanism of Vectored Interrupt.
- c. Explain the priority mechanism used for supporting nesting of interrupts. 7

Or

d. Explain how the additional capabilities required in a DMA interface are provided.

3. Answer any two of the following:

- a. Explain why Software Interrupts and Privilege Modes are required. 2 x 4
- b. What are Exceptions and how these are handled?
- c. Explain how support for debugging is provided in a processor.

4. a. What are the different forms of dependencies that lead to hazards in a pipelined processor? State the ways to deal with anyone form of the hazards.

b. What are the following? (Any three) 7

- i. Multiprocessors, 3 x 2
- ii. Multi-computers,
- iii. Data Parallel Processors,
- iv. Superscalar processors

**Section-II**  
**(20 Marks)**

5. Answer any 5 (five) of the following:

- a. Consider a processor that has 16 registers and 4 addressing modes. An operand is specified in an instruction with a combination of a register and an addressing mode. The instructions are of 16 bits in length, and these are of 4 types. Type-1 instructions specify 2 operands, Type-2 specifies 1 operand, Type-3 are branch instructions that specify an 8-bit offset and the Type-4 are without operand instructions. There are 12 Type-1 instructions, 64 Type-2, 32 Type-3 instructions. How many Type-4 instructions can be accommodated in the instruction set? 5 x 4

4

- b. A 1 Gbit SDRAM chip is organized with 4 bits per location and equal number bits for row and column addresses.
- What is the size (in bits) of the refresh buffer? 1½
  - The chip requires pins for Chip Select signal, Read/ Write' signal, Row/Column Address lines, Data Lines, Power Supply, Ground, RAS signal, CAS signal and the Clock. What will be the minimum number of pins required in the chip? 2½

- c. A computing hardware is designed with a processor generating 32bit address for memory. The areas allocated in the memory address space for I/O, EEPROM, EPROM and RAM in the physical address space are as follows:

|         |                |
|---------|----------------|
| I/O:    | 0 to (64K-1)   |
| EEPROM: | 64K to (4M-1)  |
| EPROM:  | 4M to (256M-1) |
| RAM:    | 256M to (4G-1) |

Write the Boolean expressions for the signals- I/O\_Select, EEPROM\_Select, EPROM\_Select, and RAM\_Select. 4

- d. The cache memory in a computer uses 8-way set-associative mapping. The cache size is 1MB with cache lines of 16 bytes. The memory address length is 32 bits.
- What is the set no. in cache to which the memory location 34F5AB68 (Hexadecimal) will map? 2
  - Give another memory address that will map to the same set as above. 1
  - Give one memory address that will map to the next set in cache. 1

- e. A computer system has L1 and L2 cache with access times of 5ns and 15ns respectively. Cache lines in L2 are for 256 bytes. Memory bus width is 32bits. The transfer of the first word (32 bits) of a cache line requires 50ns and the remaining words require 10ns each. The average access time of a page from online storage is 3ms. During execution a computer program experiences hit ratios of 0.95 in L1, 0.98 in L2 and 0.999 in the main memory. Determine the average memory access time during the execution of the program. 4

- f. A computer system uses a processor that generates 32-bit addresses. The MMU uses pages of 8KB size and a 2-level page table with entries of 4 bytes. The page table modules of both levels are stored in the page frames in main memory. What will be the total size of the page table? 4

- g. A 5-stage single pipeline processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The IF, ID, PO, OF and WO stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instructions, 2 clock cycles for MUL instruction, and 4 clock cycles for DIV instruction. Determine the numbers of clock cycles required for the pipelined execution for the program segment given below with use of operand forwarding- 4

| <u>Instruction</u> | <u>Meaning of instruction</u> |
|--------------------|-------------------------------|
| I1: SUB R2, R0, R1 | ; $R2 \leftarrow R0 - R1$     |
| I2: MUL R3, R2, R3 | ; $R3 \leftarrow R2 * R3$     |
| I3: DIV R4, R5, R3 | ; $R4 \leftarrow R5 / R3$     |
| I4: ADD R5, R2, R4 | ; $R5 \leftarrow R2 + R4$     |